BOUNDS ON INFORMATION-THEORETIC DIVERGENCES UNDER STRONG MUTUAL ABSOLUTE CONTINUITY

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Abstract. We derive a simple and sharp upper bound for general F-divergences between probability measures under the assumption of strong mutual absolute continuity (that is, when their density ratio is uniformly bounded on both sides). The proof relies on elementary properties of convex functions. We apply this general result to obtain explicit, algebraic bounds for several common divergences, including the chi-squared, total variation, squared Hellinger, Le Cam, and Kullback-Leibler divergences.

Introduction

Information-theoretic divergences are widely-used theoretical tools for quantifying the dissimilarities between probability measures in a systematic, reparameterization-invariant way. So-called 'F-divergences' (introduced independently by Csiszár and Ali-Silvey) are characterised by a convex 'generating function' $F : \mathbf{R}_+ \to \mathbf{R}$ satisfying $F(1) = 0$ and $F(r) > 0$ for $r \neq 1$.

Given such an F and probability measures P, Q on a common measurable space satisfying $P \ll Q$, one defines the associated divergence between P and Q by the formula

$$\mathsf{D}_F(P, Q) = \mathbf{E}_Q\left[F\left(\frac{\mathrm{d}P}{\mathrm{d}Q}\right)\right].$$

By Jensen's inequality, one sees that $\mathsf{D}_F(P, Q) \geq 0$, with equality holding if and only if $P = Q$ are indistinguishable.

Prominent examples of such divergences include the chi-squared divergence, the total variation distance, the squared Hellinger distance, the Kullback-Leibler divergence, and others. These distance measures are foundational in information theory, mathematical statistics, and machine learning, frequently appearing in sample complexity bounds, convergence rates for Markov chains, and hypothesis testing.

The purpose of this note is to demonstrate a simple and sharp upper bound for $\mathsf{D}_F(P, Q)$ in the case where P and Q are 'strongly' mutually absolutely continuous, in the sense that the density ratio

$$R = \frac{\mathrm{d}P}{\mathrm{d}Q}$$

is uniformly bounded away from both 0 and ∞. This (rather strong) condition allows for a particularly clean analysis. The proof of our main result is elementary, utilizing only that on intervals of $\mathbf{R}$, convex functions are dominated by their secants. Moreover, in specific instances, the upper bounds are seen to be surprisingly clean and user-friendly.

Main Result

The key ingredient of our bound is to dominate the convex generating function F with a linear upper bound over the restricted interval $[\alpha, \beta]$, and to subsequently evaluate the expectation which defines D_F by exploiting the normalization constraints of the density ratio.

Theorem 1. *Let $F : \mathbf{R}_+ \to \mathbf{R}$ be convex and satisfy $F(1) = 0$. Let $0 < \alpha \leq 1 \leq \beta < \infty$, and let P, Q be probability measures for which the density ratio $R = \frac{\mathrm{d}P}{\mathrm{d}Q}$ satisfies $\alpha \leq R \leq \beta$ almost everywhere. It then holds that*

$$\mathsf{D}_F(P, Q) \leq \overline{\mathsf{D}}_F(\alpha, \beta) := \frac{\beta - 1}{\beta - \alpha} F(\alpha) + \frac{1 - \alpha}{\beta - \alpha} F(\beta),$$

and this bound is unimprovable in general.

Proof. By convexity, the graph of F on the interval $[\alpha, \beta]$ lies below the secant line connecting the endpoints. That is, for any $r \in [\alpha, \beta]$, it holds that

$$F(r) \leq \frac{\beta - r}{\beta - \alpha} F(\alpha) + \frac{r - \alpha}{\beta - \alpha} F(\beta).$$

As such, we compute that

$$\begin{aligned}
\mathsf{D}_F(P, Q) &= \mathbf{E}_Q[F(R)] \\
&\leq \mathbf{E}_Q\left[\frac{\beta - R}{\beta - \alpha} F(\alpha) + \frac{R - \alpha}{\beta - \alpha} F(\beta)\right] \\
&= \frac{\beta - \mathbf{E}_Q[R]}{\beta - \alpha} F(\alpha) + \frac{\mathbf{E}_Q[R] - \alpha}{\beta - \alpha} F(\beta) \\
&= \frac{\beta - 1}{\beta - \alpha} F(\alpha) + \frac{1 - \alpha}{\beta - \alpha} F(\beta),
\end{aligned}$$

where the final equality follows noting that by the absolute continuity of P with respect to Q, the density ratio integrates to one, i.e. $\mathbf{E}_Q[R] = 1$. □

Remark 2. The condition $\alpha \leq 1 \leq \beta$ is implicitly guaranteed by the fact that R integrates to 1; unless $P = Q$ almost everywhere (forcing $R = 1$), the ratio must take values both strictly above and below 1.

Remark 3. One sees that the bound holds with exact equality when R concentrates entirely on the boundary points $\{\alpha, \beta\}$. Given the moment constraint $\mathbf{E}_Q[R] = 1$, this occurs specifically when the underlying space can be partitioned into two sets where R takes the constant values α and β, with corresponding Q-probabilities $\frac{\beta-1}{\beta-\alpha}$ and $\frac{1-\alpha}{\beta-\alpha}$, respectively.

Applications

We now apply the general bound of Theorem 1 to specific F-divergences of interest.

Example 4. For the chi-squared divergence, $F(r) = (r-1)^2$, we obtain the bound

$$\chi^2(P, Q) \leq (1 - \alpha)(\beta - 1).$$

Because $\chi^2(P, Q)$ is precisely the variance of R under Q, this inequality exactly recovers the well-known Bhatia-Davis inequality for the variance of a bounded random variable.

Example 5. For the total variation distance, $F(r) = \frac{1}{2}|r - 1|$, we obtain the bound

$$\mathsf{TV}(P, Q) \leq \frac{(1 - \alpha)(\beta - 1)}{\beta - \alpha}.$$

Example 6. For the squared Hellinger distance, $F(r) = (\sqrt{r} - 1)^2$, we obtain the bound

$$\mathsf{Hell}^2(P, Q) \leq 2\frac{(1 - \sqrt{\alpha})(\sqrt{\beta} - 1)}{\sqrt{\alpha} + \sqrt{\beta}}.$$

Example 7. For the Le Cam divergence (also known as Triangular Discrimination), $F(r) = \frac{(r-1)^2}{r+1}$, we obtain the bound

$$\Delta(P, Q) \leq 2\frac{(1 - \alpha)(\beta - 1)}{(1 + \alpha)(\beta + 1)}.$$

Example 8. While algebraic factorizations do not neatly present themselves for logarithmic divergences, the bounds remain tractable and computationally trivial. For the Kullback-Leibler divergence $F(r) = r \log r - r + 1$, we obtain the bound

$$\mathsf{KL}(P, Q) \leq \frac{\beta - 1}{\beta - \alpha} \alpha \log \alpha + \frac{1 - \alpha}{\beta - \alpha} \beta \log \beta.$$

The 'reverse' KL divergence can then be bounded directly by noting that under the prevailing assumptions, it holds that (writing $R' = \frac{\mathrm{d}Q}{\mathrm{d}P}$) $\alpha' \leq R' \leq \beta'$ with $(\alpha', \beta') = \left(\beta^{-1}, \alpha^{-1}\right)$.

Conclusion

In this note, we presented a unified and elementary method for bounding general F-divergences under the assumption of strongly bounded density ratios. By leveraging convexity and the properties of density ratios between probability measures, we established an upper bound that is sharp and easy to compute. The derivation provides immediate closed-form estimates for standard distances such as Total Variation, Chi-Squared, Hellinger, and Le Cam divergences. These bounds facilitate straightforward worst-case performance guarantees across the varied domains for which information-theoretic divergences are pertinent.